Signals & Systems (151-0575-01) Exam, February 8th, 2022 - Answer key (compact)

Condensed from the official corrected solution. 60 single-choice questions, 1 point each, all-or-nothing; calculations are not graded. Each criterion heading gives the label, name and maximum, followed by the question, the options in order (A, B, C, ...) and the key. Shared context (Box) is stated once.

Context: Consider the causal continuous-time LTI system G_c with transfer function H(s) = prod_{i=1}^{M} (s - b_i) / prod_{i=1}^{N} (s - a_i), where a_i, b_i in R. Consider the following discretization method, where poles and zeros of the continuous-time system are mapped to poles and zeros of the discretized system as follows: p_i = g(a_i Ts) and z_i = g(b_i Ts), where Ts is the sampling time and g(x) is the sigmoid function g(x) = (1 - e^(-x)) / (1 + e^(-x)). The resulting discretized transfer function is W(z) = prod_{i=1}^{M} (z - z_i) / prod_{i=1}^{N} (z - p_i).

1.a - MC question 1 - 1 points:

Question: Select the correct statement.

Options: A) The resulting discretized system is always stable | B) It is not possible to determine the stability of the system with the given information | C) None of the other statements is correct | D) If the continuous-time system is stable, then the discretized system is unstable | E) If the continuous-time system is unstable, then the discretized system is unstable | F) The resulting discretized system is always unstable

Key: option A - The resulting discretized system is always stable

2.a - MC question 2 - 1 points:

Question: Select the correct statement.

Options: A) It is not possible to determine time-invariance of the system with the given information | B) The resulting discretized system is always non-causal | C) None of the other statements is correct | D) The resulting discretized system is always nonlinear | E) The resulting discretized system is always time-invariant

Key: option E - The resulting discretized system is always time-invariant

3.a - MC question 3 - 1 points:

Question: The continuous-time system dq/dt(t) = A_c q(t) + B_c u(t) is discretized, resulting in the discrete-time system q[n+1] = A_d q[n] + B_d u[n]. Mark the true statement.

Options: A) In Exact discretization, when Ts -> 0 then the input matrix B_d -> I | B) In Euler discretization, when Ts -> 0 then the state matrix A_d -> 0 | C) None of the other statements is correct | D) In both Euler and Exact discretization, when Ts -> 0 then the state matrix A_d -> I

Key: option D - In both Euler and Exact discretization, when Ts -> 0 then the state matrix A_d -> I

Context: Consider the continuous-time LTI system G_c with transfer function H(s) = 1 / (s - 0.5). G_c is discretized using Euler discretization with sampling time Ts.

4.a - MC question 4 - 1 points:

Question: What is the ratio k_r between the CT DC gain and the DT DC gain? Hint: the DC gain is the gain at zero CT frequency, i.e. omega = 0.

Options: A) k_r = exp(-Ts) | B) k_r = -Ts | C) k_r = exp(Ts) | D) k_r = 1 | E) k_r = 0 | F) k_r = Ts

Key: option D - k_r = 1

5.a - MC question 5 - 1 points:

Question: For which sampling times Ts is the discretized system unstable?

Options: A) 0 <= Ts <= 0.5 | B) 0 <= Ts <= 2 | C) None of the other statements is correct | D) Ts >= 1 | E) Ts >= 0 | F) Ts >= 2 | G) Ts >= 0.5 | H) 0 <= Ts <= 4

Key: option E - Ts >= 0

Context: Consider the continuous-time signals x_1(t) = cos(w_1 t) and x_2(t) = sin(w_2 t), with w_1, w_2 in R, given as plots for t = 0..11: x_1(t) starts at 1 for t = 0, has minima at t = 2.5 and 7.5 and maxima at t = 5 and 10; x_2(t) starts at 0 for t = 0, has maxima at t = 1.5 and 7.5, zero crossings at t = 3, 6 and 9, and a minimum at t = 4.5. The signals are sampled with sampling period Ts = 0.1, resulting in the discrete-time signals x_1[n] = cos(Omega_1 n) and x_2[n] = sin(Omega_2 n).

6.a - MC question 6 - 1 points:

Question: What is the discrete-time frequency Omega_1?

Options: A) Omega_1 = pi/2 | B) Omega_1 = 5pi/6 | C) Omega_1 = 1/50 | D) Omega_1 = pi/25 | E) Omega_1 = 2/3 | F) Omega_1 = pi

Key: option D - Omega_1 = pi/25

7.a - MC question 7 - 1 points:

Question: What is the period N_2 of the discrete-time signal x_2[n]?

Options: A) N_2 = 60 | B) N_2 = 100 | C) N_2 = 20 | D) N_2 = 120 | E) N_2 = 70 | F) N_2 = 30

Key: option A - N_2 = 60

8.a - MC question 8 - 1 points:

Question: What is the fundamental period N of the discrete-time signal x[n] = cos(pi/7 n) + sin(pi/8 n)?

Options: A) N = 30 | B) N = 56 | C) N = 84 | D) N = 132 | E) N = 112 | F) N = 24

Key: option E - N = 112

Context: Consider the following discrete-time systems with input u and output y:
G_A: q[n+1] = 0.7 q[n] + 0.5 u[n], y[n] = q[n]
G_B: q[n+1] = 2 pi q[n] + 0.5 u[n], y[n] = 5 q[n]
G_C: q[n+1] = -0.9 q[n] + 2 u[n+1], y[n] = 0.5 q[n]
G_D: q[n+1] = 3^n q[n] + 0.5 u[n+1], y[n] = 0.5 q[n]

9.a - MC question 9 - 1 points:

Question: Select the answer containing all linear systems.

Options: A) G_A | B) G_B | C) G_C | D) G_D | E) G_A, G_B | F) G_A, G_C | G) G_A, G_D | H) G_B, G_C | I) G_B, G_D | J) G_C, G_D | K) G_A, G_B, G_C | L) G_A, G_B, G_D | M) G_A, G_C, G_D | N) G_B, G_B, G_D | O) G_A, G_B, G_C, G_D

Key: option O - G_A, G_B, G_C, G_D

10.a - MC question 10 - 1 points:

Question: Select the answer containing all time-invariant systems.

Options: A) G_A | B) G_B | C) G_C | D) G_D | E) G_A, G_B | F) G_A, G_C | G) G_A, G_D | H) G_B, G_C | I) G_B, G_D | J) G_C, G_D | K) G_A, G_B, G_C | L) G_A, G_B, G_D | M) G_A, G_C, G_D | N) G_B, G_B, G_D | O) G_A, G_B, G_C, G_D

Key: option K - G_A, G_B, G_C

11.a - MC question 11 - 1 points:

Question: Select the answer containing all causal systems.

Options: A) G_A | B) G_B | C) G_C | D) G_D | E) G_A, G_B | F) G_A, G_C | G) G_A, G_D | H) G_B, G_C | I) G_B, G_D | J) G_C, G_D | K) G_A, G_B, G_C | L) G_A, G_B, G_D | M) G_A, G_C, G_D | N) G_B, G_B, G_D | O) G_A, G_B, G_C, G_D

Key: option E - G_A, G_B

12.a - MC question 12 - 1 points:

Question: Select the answer containing all BIBO LTI stable systems.

Options: A) G_A | B) G_B | C) G_C | D) G_D | E) G_A, G_B | F) G_A, G_C | G) G_A, G_D | H) G_B, G_C | I) G_B, G_D | J) G_C, G_D | K) G_A, G_B, G_C | L) G_A, G_B, G_D | M) G_A, G_C, G_D | N) G_B, G_B, G_D | O) G_A, G_B, G_C, G_D

Key: option A - G_A

Context: Consider the following LTI system: two identical copies of the LTI system G, with inputs u_1 and u_2 and outputs y_1 and y_2, whose outputs are summed: y = y_1 + y_2. The inputs are u_1 = s[n] and u_2 = 2 s[n], with s[n] being the unit step. The observed output is y[-1] = 0, y[0] = 3, y[1] = 1, y[2] = 3, y[3] = 0 (and y[n] = 0 for all other n).

13.a - MC question 13 - 1 points:

Question: What is the impulse response of G? (The value marked (n=0) is at time n = 0.)

Options: A) {h[n]} = {..., 0, 2, 1 (n=0), 1/3, 1/4, 0, 0, ...} | B) None of the other statements is correct | C) {h[n]} = {..., 0, 2 (n=0), 2, 0, ...} | D) {h[n]} = {..., 0, 1 (n=0), 2, 2, 2, 0, ...} | E) {h[n]} = {..., 0, 1 (n=0), 1, 0, ...} | F) {h[n]} = {..., 0, 1 (n=0), -2/3, 2/3, 0, 0, ...}

Key: option F - {h[n]} = {..., 0, 1 (n=0), -2/3, 2/3, 0, 0, ...}

14.a - MC question 14 - 1 points:

Question: Select the correct statement.

Options: A) The system G is non-causal and unstable. | B) None of the other statements is correct | C) The system G is non-causal and stable. | D) It is not possible to determine if the system G is stable. | E) The system G is causal and stable. | F) The system G is causal and unstable.

Key: option E - The system G is causal and stable.

Context: Let G be an LTI system of the form q[n+1] = a q[n] + b u[n], y[n] = c q[n] + d u[n], with a, b, c, d in R, and q[0] = 0. We can measure the output of the step response at the following time instants: r[0] = 1, r[1] = 2, r[2] = 3.1, and we know that the system's impulse response is zero for n < 0.

15.a - MC question 15 - 1 points:

Question: Select the correct statement.

Options: A) None of the other statements is correct | B) The system G is unstable and causal | C) The system G is stable and causal | D) It is not possible to determine stability and causality with the given information | E) The system G is stable and non-causal | F) The system G is unstable and non-causal

Key: option B - The system G is unstable and causal

16.a - MC question 16 - 1 points:

Question: Given that q[1] = 0.5, select the correct statement.

Options: A) None of the other statements is correct | B) The output matrix of the system G is c = 3 | C) It is not possible to determine the system's input and output matrices b and c with the given information | D) The input matrix of the system G is b = 4 | E) The output matrix of the system G is c = -1 | F) The input matrix of the system G is b = -1 | G) The output matrix of the system G is c = 2

Key: option G - The output matrix of the system G is c = 2

17.a - MC question 17 - 1 points:

Question: Consider the system G consisting of two systems G_1 and G_2 in parallel: the input u is applied to both, and their outputs y_1 and y_2 are summed to give y = y_1 + y_2. Select the correct statement.

Options: A) None of the other statements is correct | B) The system G is stable if and only if G_1 is unstable and G_2 is stable | C) The system G is stable if and only if G_1 is stable and G_2 is stable | D) The system G is stable if and only if G_1 is unstable and G_2 is unstable

Key: option A - None of the other statements is correct

Context: Consider the discrete-time LTI system with impulse response h[n] = 0 for n <= 0 and h[n] = a^n - b^n for n > 0, with a, b in R, a != 0, b != 0.

18.a - MC question 18 - 1 points:

Question: For which z in C does the z-transform H(z) = sum_{n=-inf}^{inf} h[n] z^(-n) exist?

Options: A) |z| < |a|, |z| < |b| | B) |z| > |a|, |z| < |b| | C) |z| < |a|, |z| > |b| | D) |z| <= |a|, |z| <= |b| | E) |z| >= |a|, |z| >= |b| | F) |z| > |a|, |z| > |b|

Key: option F - |z| > |a|, |z| > |b|

19.a - MC question 19 - 1 points:

Question: For which values of a and b is the output of the system always y[n] = 0?

Options: A) a > b | B) a < b | C) a > b + 1 | D) a = b - 1 | E) a = b | F) a = b + 1

Key: option E - a = b

20.a - MC question 20 - 1 points:

Question: Let a = 2b. If the input to the system is u[n] = 1^n, find b such that the output y[n] = 1 for all n.

Options: A) b = +/- sqrt(2) | B) b = 1 +/- sqrt(2) | C) b = +/- 1 | D) b = +/- sqrt(2)/2 | E) b = 1 +/- sqrt(2)/2 | F) b = +/- 1/2

Key: option E - b = 1 +/- sqrt(2)/2

Context: Consider the causal, discrete-time LTI systems G_1 and G_2. G_3 is the series (cascade) connection u_3 -> G_1 -> G_2 -> y_3. G_4 is the parallel connection: u_4 is applied to G_1 and G_2 and their outputs are summed to give y_4. h_i[n] denotes the impulse response of system G_i, for i = 1, 2, 3, 4.

21.a - MC question 21 - 1 points:

Question: Select the correct statement.

Options: A) G_3 always has a finite impulse response | B) None of the other statements is correct | C) G_3 always has an infinite impulse response | D) If G_1 and G_2 have a finite impulse response then G_3 has a finite impulse response

Key: option D - If G_1 and G_2 have a finite impulse response then G_3 has a finite impulse response

22.a - MC question 22 - 1 points:

Question: Assume that h_1[n] = 1 for all n >= 0, and that we know h_4[0] = 2. Which of these values corresponds to h_3[0]?

Options: A) h_3[0] = 4 | B) h_3[0] = 3 | C) h_3[0] = 1 | D) It is not possible to determine h_3[0] with the given information | E) h_3[0] = 2

Key: option C - h_3[0] = 1

23.a - MC question 23 - 1 points:

Question: You are given the step responses of systems G_3 and G_4: r_3[n] = {..., 0, 0, 0 (n=0), 2, ...} and r_4[n] = {..., 0, 0, 1 (n=0), 2, ...}. Furthermore, the impulse responses of G_1 and G_2 have the forms h_1[n] = {..., 0, 0, 0 (n=0), a, ...} and h_2[n] = {..., 0, 0, b (n=0), c, ...}, with a > b. Determine a and b.

Options: A) a = 3, b = -1, and c = 0 | B) a = 5, b = 0, and c = 3 | C) a = 1, b = 0, and c = -2 | D) a = 2, b = 1, and c = 4 | E) a = 2, b = 1, and c = -1 | F) a = 1, b = 0, and c = 1 | G) a = 3, b = 2, and c = 2 | H) a = 4, b = -1, and c = 0

Key: option E - a = 2, b = 1, and c = -1

Context: Consider the causal, discrete-time LTI system G with state-space representation q[n+1] = a q[n] + b u[n], y[n] = c q[n] + d u[n], where a, b, c, d in R. The associated transfer function is H(z) = z / (z - 0.5). The input to the system is u[n] = (3/2)^n, and the corresponding state and output are q[n] = (1/2)(3/2)^n and y[n] = (3/2)^(n+1), respectively.

24.a - MC question 24 - 1 points:

Question: Determine the transfer function W(z) from q[n] to y[n] as a function of a, b, c, d.

Options: A) W(z) = (dz + cb - da) / b | B) W(z) = (bz + c - da) / d | C) W(z) = (cz - da) / b | D) W(z) = (dz + cb - da) / (z - a) | E) W(z) = (dz + cb - da) / z | F) W(z) = ((d + a)z - cb) / b

Key: option A - W(z) = (dz + cb - da) / b

25.a - MC question 25 - 1 points:

Question: Determine the values a, b, c, d.

Options: A) a = 0.5, b = 0.5, c = 1, d = 1 | B) a = 1, b = 0.5, c = 1, d = 0.5 | C) a = 0.5, b = 3, c = 1, d = 0 | D) a = 0.5, b = 0, c = 1, d = 2 | E) a = 0.8, b = 2, c = 1, d = 0.5 | F) a = 0.5, b = 1, c = 0.5, d = 1

Key: option A - a = 0.5, b = 0.5, c = 1, d = 1

26.a - MC question 26 - 1 points:

Question: The largest ROC of a transfer function H(z) is 0.5 < |z| < 1.5. Determine which of four pole-zero plots (Plot A to Plot D; poles x, zeros o, unit circle shown) represents H(z). [Figure not included.]

Options: A) Plot A | B) Plot B | C) Plot C | D) Plot D

Key: option A - Plot A

Context: Consider the signal x(t) = sgn(sin(4 pi t)), where sgn(x) = 1 if x >= 0 and 0 otherwise.

27.a - MC question 27 - 1 points:

Question: For which of the following sampling times Ts will the signal x(t) experience aliasing?

Options: A) None of the others answers is correct | B) Ts = 0.01 | C) Ts = 0.1 | D) Ts = 0.05 | E) Ts = 0.2 | F) Ts = 0.5

Key: option F - Ts = 0.5

28.a - MC question 28 - 1 points:

Question: If the signal is sampled with sampling time Ts = 0.001, what is the fundamental period N?

Options: A) N = 10 | B) None of the others answers is correct | C) N = 50 | D) N = 500 | E) N = 5000 | F) N = 1000 | G) N = 5 | H) N = 100

Key: option D - N = 500

Context: Six plots (Plot A to Plot F) show the magnitude of the DFT coefficients of different DT signals, computed using a DFT length of 20 (k = 0..19). [Figure not included.]

29.a - MC question 29 - 1 points:

Question: Let {x_1[n]} = {cos(4pi/9 n)}. Which plot shows the magnitude of {X_1[k]}?

Options: A) Plot A | B) Plot B | C) Plot C | D) Plot D | E) Plot E | F) Plot F

Key: option D - Plot D

30.a - MC question 30 - 1 points:

Question: Let {x_2[n]} = {0.5 e^(j 3pi n/9) - 0.5 e^(j 2pi n/3)}. Which plot shows the magnitude of {X_2[k]}?

Options: A) Plot A | B) Plot B | C) Plot C | D) Plot D | E) Plot E | F) Plot F

Key: option B - Plot B

31.a - MC question 31 - 1 points:

Question: Let {x_3[n]} = {alpha 1^n}, where alpha in R. Which plot shows the magnitude of {X_3[k]}?

Options: A) Plot A | B) Plot B | C) Plot C | D) Plot D | E) Plot E | F) Plot F

Key: option F - Plot F

32.a - MC question 32 - 1 points:

Question: The DT Fourier transform of the output of a system G is Y(Omega) = 1 + e^(-j Omega). To what signal does Y(Omega) correspond in the time domain?

Options: A) y[n] = s[n], where s[n] is the step signal | B) None of the others answers is correct | C) y[n] = 2 delta[n-2] | D) y[n] = 2 sin[n] | E) y[n] = sin[n-1] | F) y[n] = delta[n] + 1 | G) y[n] = delta[n+2] + delta[n] | H) y[n] = cos[n+1] | I) y[n] = 2 cos[n] | J) y[n] = delta[n] + delta[n-1]

Key: option J - y[n] = delta[n] + delta[n-1]

33.a - MC question 33 - 1 points:

Question: Consider the magnitude of the DT Fourier transform |X(Omega)| of the real sequence {x[n]}: it consists only of impulses (Dirac deltas) at Omega = pi/3, 2pi/3, pi, 4pi/3 and 5pi/3 and is zero elsewhere. Which of the following statements is correct?

Options: A) The sequence {x[n]} is not absolutely summable. | B) The sequence {x[n]} has a fundamental period of N = 4. | C) sum_{n=-inf}^{inf} x[n] = 4. | D) None of the others answers is correct | E) The sequence {x[n]} has a fundamental period of N = 8.

Key: option A - The sequence {x[n]} is not absolutely summable.

Context: Consider an LTI system G with transfer function H(z) = (z^2 + 1) / z^2. Consider also two input signals u_1[n] and u_2[n] with DFS coefficients U_1[k] = {0, 2} and U_2[k] = {0, -2j, 0, 2j}, respectively.

34.a - MC question 34 - 1 points:

Question: Determine the DFS coefficients of u_3[n] = u_1[n] + u_2[n].

Options: A) {U_3[k]} = {0, 2j, 2, -2j} | B) {U_3[k]} = {2, 2, -2j, 2} | C) {U_3[k]} = {2, -j, 1, j} | D) {U_3[k]} = {2, -2j, 2, 2j} | E) {U_3[k]} = {0, -2j, 2, 2j} | F) {U_3[k]} = {0, -2j, 4, 2j}

Key: option F - {U_3[k]} = {0, -2j, 4, 2j}

35.a - MC question 35 - 1 points:

Question: Assume that u_3[n] = alpha_1 u_1[n] + alpha_2 u_2[n], and that the periodic signal u_3[n] = {1, 2, 1, -4}. Determine alpha_1 and alpha_2.

Options: A) alpha_1 = 8, alpha_2 = 6 | B) alpha_1 = 8, alpha_2 = 3 | C) alpha_1 = 1, alpha_2 = 3 | D) alpha_1 = 2, alpha_2 = 6 | E) alpha_1 = 2, alpha_2 = 3 | F) alpha_1 = 2, alpha_2 = 4

Key: option C - alpha_1 = 1, alpha_2 = 3

36.a - MC question 36 - 1 points:

Question: Determine the DFS coefficients of y_3[n] = G(u_1[n] + u_2[n]).

Options: A) {Y_3[k]} = {0, -2j, 8, 2j} | B) {Y_3[k]} = {0, 0, 8, 0} | C) {Y_3[k]} = {0, 0, 16, 0} | D) {Y_3[k]} = {2, 2j, 8, 2j} | E) {Y_3[k]} = {2, 4j, 0, -4j} | F) {Y_3[k]} = {2, -4j, 0, 4j}

Key: option B - {Y_3[k]} = {0, 0, 8, 0}

37.a - MC question 37 - 1 points:

Question: Which of four magnitude/phase plots (Plot A to Plot D) represents the frequency response H(Omega)? [Figure not included.]

Options: A) Plot A | B) Plot B | C) Plot C | D) Plot D

Key: option C - Plot C

38.a - MC question 38 - 1 points:

Question: If the input signal is u[n] = 2 + cos(pi/2 n), which of the following corresponds to the output signal y[n]?

Options: A) y[n] = 2 + cos(pi/2 n - pi/2) | B) y[n] = 0 | C) y[n] = 2 | D) y[n] = 4 | E) y[n] = 2 + sin(pi/2 n - pi/2) | F) y[n] = 1 + cos(pi/2 n)

Key: option C - y[n] = 2

Context: Consider causal LTI systems G_1 and G_2, where G_3 is the cascade u -> G_2 -> G_1 -> y. The input u[n] is applied both to G_3 and, separately, to G_1, whose output is w[n]. The output of G_3 is y[n] = delta[n], and the output of G_1 is w[n] = (1/2)^n.

39.a - MC question 39 - 1 points:

Question: Which of four magnitude/phase plots (Plot A to Plot D) shows the frequency response H_2(Omega) of system G_2? [Figure not included.]

Options: A) Plot A | B) Plot B | C) Plot C | D) Plot D

Key: option B - Plot B

40.a - MC question 40 - 1 points:

Question: Assume that U(z) = 1 - 0.25 z^(-1). Which of four magnitude/phase plots (Plot A to Plot D) shows the frequency response H_3(Omega) of system G_3? [Figure not included.]

Options: A) Plot A | B) Plot B | C) Plot C | D) Plot D

Key: option C - Plot C

Context: Let y_1[n] = |x_1[n]| + |x_2[n]| and y_2[n] = |x_1[n] + x_2[n]|. Let {Y_1[k]} and {Y_2[k]} be the DFT coefficients of the sequences {y_1[n]} and {y_2[n]}, respectively, both of length N.

41.a - MC question 41 - 1 points:

Question: Which statement is true?

Options: A) Y_1[0] >= Y_2[0] | B) Y_1[0] = Y_2[0] | C) Y_1[0] <= Y_2[0] | D) None of the other statements can be determined. | E) Y_1[0] < Y_2[0] | F) Y_1[0] > Y_2[0]

Key: option A - Y_1[0] >= Y_2[0]

42.a - MC question 42 - 1 points:

Question: Assume x_1[n] = (1)^n and x_2[n] = cos(pi n) are sequences of length N = 2. Which statement is true?

Options: A) sum_{k=0}^{1} Y_1[k] < sum_{k=0}^{1} Y_2[k] | B) sum_{k=0}^{1} Y_1[k] <= sum_{k=0}^{1} Y_2[k] | C) sum_{k=0}^{1} Y_1[k] >= sum_{k=0}^{1} Y_2[k] | D) sum_{k=0}^{1} Y_1[k] = sum_{k=0}^{1} Y_2[k] | E) sum_{k=0}^{1} Y_1[k] > sum_{k=0}^{1} Y_2[k] | F) None of the other statements can be determined.

Key: option D - sum_{k=0}^{1} Y_1[k] = sum_{k=0}^{1} Y_2[k]

Context: Consider the following discrete-time LTI filters:
G_A: H(z) = w^2 / (z^2 + z (sqrt(2) w - 2) + (1 - sqrt(2) w + w^2))
G_B: H(z) = w^2 / (z^4 - z^2 (sqrt(2) w - 2) + (1 - sqrt(2) w + w^2))
G_C: H(z) = w^2 / (z^2 - z (sqrt(2) w - 2) + (1 - sqrt(2) w + w^2))
G_D: H(z) = w^2 / (-z^2 - z (sqrt(2) w - 2) + (1 - sqrt(2) w + w^2))

43.a - MC question 43 - 1 points:

Question: Which of the following is a low-pass filter?

Options: A) G_A | B) G_B | C) G_C | D) G_D | E) None of the others

Key: option A - G_A

44.a - MC question 44 - 1 points:

Question: Which of the following is a high-pass filter?

Options: A) G_A | B) G_B | C) G_C | D) G_D | E) None of the others

Key: option C - G_C

45.a - MC question 45 - 1 points:

Question: Which of the following is a band-pass filter?

Options: A) G_A | B) G_B | C) G_C | D) G_D | E) None of the others

Key: option B - G_B

46.a - MC question 46 - 1 points:

Question: Consider a transfer function represented by its pole-zero plot: the pole(s) are at the origin z = 0 and there are two zeros on the positive real axis inside the unit circle, at about z = 0.7 and z = 0.9. What kind of filter does it implement?

Options: A) Notch filter | B) Band-pass filter | C) Band-stop filter | D) High-pass filter | E) None of these | F) Low-pass filter

Key: option D - High-pass filter

47.a - MC question 47 - 1 points:

Question: Consider a discrete-time frequency transformation that maps the discrete-time frequency Omega to Omega_t: Omega = 0 is mapped to Omega_t = 0, Omega_t rises smoothly to about 0.9 pi at Omega = pi/2, and falls back symmetrically to Omega_t = 0 at Omega = pi. If this transformation is applied to a low-pass filter, what kind of filter is obtained?

Options: A) Band-stop filter | B) Low-pass filter | C) Band-pass filter | D) High-pass filter | E) Notch filter | F) None of these

Key: option A - Band-stop filter

Context: A continuous-time signal x(t) is sampled with sampling time Ts = 0.001. The DFT of the sampled signal x[n], computed over 101 samples (k = 0..100), has clear peaks at k = 10 (|X| about 5), k = 30 (about 9), k = 40 and 41 (about 7 and 6), k = 60 and 61 (about 6 and 7), k = 71 (about 9) and k = 91 (about 5); between k = 30 and k = 71 the magnitude stays around 1, elsewhere it is close to 0.

48.a - MC question 48 - 1 points:

Question: Which DFT coefficient would you set to zero in order to attenuate a disturbance at 100Hz?

Options: A) k = 10 and k = 31 | B) k = 10 and k = 91 | C) k = 50 | D) k = 40 | E) k = 5 and k = 96 | F) k = 5 and k = 11

Key: option B - k = 10 and k = 91

49.a - MC question 49 - 1 points:

Question: Which DFT coefficients would you set to zero in order to filter all the components with frequency higher than 200Hz?

Options: A) k <= 20 | B) k >= 30 | C) k >= 30 and k <= 71 | D) k <= 40 | E) k >= 40 | F) k >= 20 and k <= 81

Key: option F - k >= 20 and k <= 81

50.a - MC question 50 - 1 points:

Question: Consider a continuous-time signal x(t) = sin(2 pi f t), that is sampled with sampling time Ts resulting in the discrete-time signal x[n] = sin(2 pi f Ts n). Mark the correct statement.

Options: A) Even if x[n] didn't experience aliasing then up-sampling x[n] can cause aliasing | B) If f in N and Ts in N then x[n] will not be periodic | C) If f = 30 then Ts >= 0.3 will avoid aliasing | D) Even if x[n] didn't experience aliasing then down-sampling x[n] can cause aliasing | E) None of the others

Key: option D - Even if x[n] didn't experience aliasing then down-sampling x[n] can cause aliasing

Context: Consider periodic input signals u_1[n] and u_2[n] with DFS coefficients U_1[k] = {3, 1.5, 1.5} (k = 0, 1, 2) and U_2[k] = {5, 2.5, 0, 0, 2.5} (k = 0..4), with U_1[k], U_2[k] in R for all k.

51.a - MC question 51 - 1 points:

Question: Determine the DFS coefficients Y_1[k] of the output, after applying a filter with impulse response h[n] = 1/3 for n = 0, 1, 2 and h[n] = 0 otherwise to the input u_1[n].

Options: A) Y_1[k] = {3, 0, 0} | B) Y_1[k] = {0, 1.5, 1.5} | C) Y_1[k] = {0, 0, 0} | D) Y_1[k] = {0, 1.5j, -1.5j} | E) Y_1[k] = {3, 0, 1.5} | F) None of the others

Key: option A - Y_1[k] = {3, 0, 0}

52.a - MC question 52 - 1 points:

Question: Determine the output y_2[n], after applying a filter with impulse response h[-1] = 1, h[0] = 2, h[1] = 1 and h[n] = 0 otherwise to the input u_2[n]. Note that 0 < alpha < 1 and beta > 1.

Options: A) y_2[n] = alpha | B) y_2[n] = 1 | C) y_2[n] = 1 + alpha cos(pi/5) | D) y_2[n] = 1 + alpha cos(2pi/5 + pi/5) | E) y_2[n] = 1 + alpha cos(2pi/5) | F) y_2[n] = beta + alpha sin(pi/5)

Key: option E - y_2[n] = 1 + alpha cos(2pi/5)

53.a - MC question 53 - 1 points:

Question: Determine the output y_1[n], after applying a filter with impulse response h[n] = 0.2 for n = -2, -1, 0, 1, 2 and h[n] = 0 otherwise to the input u_1[n]. Note that 0 < alpha < 1 and beta > 1. (The options are written as y_2[n] in the original exam.)

Options: A) y_2[n] = 1 | B) y_2[n] = 1 + alpha cos(pi/3 + pi) | C) y_2[n] = 1 + alpha cos(2pi/3 + pi) | D) y_2[n] = 1 + alpha sin(2pi/3) | E) y_2[n] = 1 + beta sin(2pi/3 + pi) | F) y_2[n] = 1 + beta cos(pi/3)

Key: option C - y_2[n] = 1 + alpha cos(2pi/3 + pi)

Context: Consider the first-order continuous-time high-pass filter H(s) = 1 / (1 + w_c s^(-1)) (1), where w_c is the cut-off frequency. The filter is discretized using the bilinear transform with sampling time Ts = 1, obtaining a first-order discrete-time high-pass filter H_hp(z) with pole z_p. Furthermore, H_bp(z) denotes a band-pass filter obtained via a frequency transformation f(z), such that H_bp(z) = H_hp(f(z)).

54.a - MC question 54 - 1 points:

Question: Determine w_c such that z_p = 0.5.

Options: A) w_c = 1 | B) w_c = 2 | C) w_c = 8 | D) w_c = 6 | E) w_c = 10 | F) w_c = 3

Key: option D - w_c = 6

55.a - MC question 55 - 1 points:

Question: Determine f(z).

Options: A) f(z) = -z | B) f(z) = z | C) f(z) = z^3 | D) f(z) = z^2 | E) f(z) = -z^3 | F) f(z) = -z^2

Key: option D - f(z) = z^2

56.a - MC question 56 - 1 points:

Question: Determine the desired cut-off frequencies of H_bp(z), denoted as Omega_0^des and Omega_1^des, as functions of z_p. Hint: the desired cut-off frequency Omega_c^des of H_hp(z) is obtained after pre-warping w_c.

Options: A) Omega_0^des = (1 + z_p) / (1 - z_p), and Omega_1^des = (1 + z_p) / (1 - z_p) + pi | B) None of the others | C) Omega_0^des = arctan z_p, and Omega_1^des = arctan z_p + pi | D) Omega_0^des = 1 + z_p, and Omega_1^des = 1 - z_p + pi | E) Omega_0^des = arctan((1 + z_p) / (1 - z_p)), and Omega_1^des = arctan((1 + z_p) / (1 - z_p)) + pi

Key: option E - Omega_0^des = arctan((1 + z_p) / (1 - z_p)), and Omega_1^des = arctan((1 + z_p) / (1 - z_p)) + pi

Context: Consider a band-stop filter with cut-off frequencies w_0 and w_1, given by its Bode magnitude |H_BP(w)| in dB for w from 10^0 to 10^2 rad/s: the magnitude is 0 dB at low and high frequencies and has a deep notch between them; it is about -3 dB at w = 5 rad/s and at w = 10 rad/s.

57.a - MC question 57 - 1 points:

Question: Which frequency mapping f(s) would you choose for generating the band-stop filter above from a second-order low-pass Butterworth filter?

Options: A) f(s) = s^2 / (s^2 + w_0 w_1) | B) None of the others | C) f(s) = (s^2 + w_0 w_1) / s | D) f(s) = s / (s^2 + w_0 w_1) | E) f(s) = s / (s^2 + (w_0 w_1)^2)

Key: option D - f(s) = s / (s^2 + w_0 w_1)

58.a - MC question 58 - 1 points:

Question: Which of the following second-order low-pass Butterworth filters could have been used to generate the band-stop filter above?

Options: A) H_LP(s) = 0.04 / (s^2 + 0.2828s + 0.04) | B) H_LP(s) = 4 / (s^2 + 2.828s + 4) | C) H_LP(s) = 0.01 / (s^2 + 0.1414s + 0.01) | D) H_LP(s) = 1 / (s^2 + 1.414s + 1) | E) H_LP(s) = 25 / (s^2 + 7.071s + 25)

Key: option A - H_LP(s) = 0.04 / (s^2 + 0.2828s + 0.04)

59.a - MC question 59 - 1 points:

Question: Consider the discrete-time system H(z) = 1 / (z (z - a)), where a in R is a non-negative, but unknown constant. To identify the system, you input u[n] = sin(pi/2 n). The output signal y[n] is equal to (1/sqrt(5)) sin(pi/2 n + atan(-2)). Assuming that there is no noise in the measurement, what is the value of a?

Options: A) a = 2 | B) a = 0.1 | C) It is not possible to identify the system with this information | D) a = 1 | E) None of the others | F) a = 0.5

Key: option A - a = 2

60.a - MC question 60 - 1 points:

Question: Consider the following experimental procedure for estimating the frequency response H_hat(Omega) of a given discrete-time LTI system. Assume that the effect of the transient is negligible since the length of the experiment N = 100 is long enough. The frequency response estimate is measured at frequencies Omega_l = 2 pi l / N, by applying an input u[n] = A cos(Omega_l n). The table summarizes the DFT coefficients of the output measurements |Y_m[l]| and the corresponding frequency response estimates |H_hat[Omega_l]|; for all measurements, the DFT coefficients have zero phase.
l: 0, 5, 10, 15, 20, 25, 30, 35, 40, 45
|Y_m[l]|: 10, 9.1, 7.45, 6.05, 5.1, 4.4, 3.95, 3.65, 3.45, 3.35
|H_hat[Omega_l]|: 2, 1.82, 1.49, 1.21, 1.02, 0.88, 0.79, 0.73, 0.69, 0.67
What is the amplitude A of the applied sinusoidal input?

Options: A) A = 0.5 | B) A = 0.1 | C) A = 5 | D) A = 0.2 | E) A = 2 | F) A = 10

Key: option B - A = 0.1